

ZAID HANIF

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EDUCATION

West Virginia University, Morgantown, WV

August 2023–May 2027

Bachelor of Science in Computer Science / Minor in Physics

GPA: 3.63/4.0

Relevant Coursework: Operating Systems, Networking, Computer Organization, Discrete Math, Digital Logic Design, Data Structures and Algorithms, Software Engineering, Artificial Intelligence

EXPERIENCE

Southwest Research Institute (SwRI), San Antonio, TX

May 2026-July 2026

Software Engineering Intern

- Developed a Python RAG LLM pipeline and semantic search prototype using Langchain that chunks documentation, generates local embeddings, and returns top-k sources for user queries.
- Modeled documents, chunks, and vector embeddings with SQLite, sqlite-vec, and SQLAlchemy 2.0, adding checksum-based deduplication.
- Refactored ingestion, storage, and query workflows into a modular QueryAPI with dataclasses, structured logging, and typed vector serialization.
- Containerized the application with Docker and added Pydantic-based environment/CLI configuration plus Ollama-based answer generation.

PROJECTS

MPX Operating System

C, x86 Assembly

- Developed a bare-metal x86 operating system **kernel** in a freestanding environment, bootstrapping the GDT, IDT, PIC, and virtual memory paging with modeled RAM.
- Implemented cooperative multitasking through a custom x86 assembly ISR saving/restoring **16** registers per context switch, with **priority-based scheduling**.
- Built a first-fit dynamic heap allocator using **Memory Control Blocks**, supporting block splitting on allocation and adjacent-block merging on free to **minimize external fragmentation**.
- Engineered an interrupt-driven serial I/O driver for the Intel 8250 UART, managing Device and I/O Control Blocks with asynchronous read/write operations and ring-buffered input.

Momentum - Fitness Tracker

React (TypeScript), Express

- Built a workout tracker with a **React/TypeScript** frontend and an **Express** REST API connected to a **MySQL** database over SSL.
- Implemented token-based authentication with short-lived access tokens and 90-day refresh tokens, tracking active sessions server-side and revoking them on logout.
- Wrote a streak algorithm that converts workout timestamps into unique calendar dates and counts consecutive days backward to determine the user's current activity streak.
- Led an 8 member software engineering team, breaking down requirements into scoped tasks according to Agile methodology.

ByteBistro - AI Powered Recipe Assistant

React (JavaScript), Express

- Developed a full-stack application integrating **Google Gemini** for recipe generation and **Hugging Face Stable Diffusion** for food illustrations, with **JWT** authentication and **PostgreSQL** data persistence.
- Implemented a caching system for generated images using Supabase Storage.
- Built responsive **React** frontend with Framer Motion animations, theme switching, and accessibility features, deployed on Render with environment-based configuration.

TECHNICAL SKILLS

- Languages: C++, C, Python, JavaScript, TypeScript, Java, MIPS Assembly, HTML, CSS, SQL
- Frameworks/Tools: Git, GitHub, React, Vite, Express.js, SFML, TailwindCSS, Supabase

STUDENT ORGANIZATIONS

Society of Hispanic Professional Engineers (SHPE)

- Managed chapter finances and budget allocation as Treasurer
- Represented WVU SHPE at the 2025 National Convention in Philadelphia